

# Manual (de)initialization

A Rust and C++ showcase

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```
1 trait Exein: Rust + Security {}
```

# About

# Boring code



```
1 pub struct Foo {
2     name: String,
3     data: Vec<i32>,
4     len: usize,
5 }
6
7 impl Foo {
8     pub fn new(name: &str) -> Self {
9         let name = name.to_owned();
10        let len = name.len();
11
12        Self {
13            name,
14            data: Vec::new(),
15            len,
16        }
17    }
18 }
19
20 fn main() {
21     let foo = Foo::new("Rustikon");
22 }
```

```
1 #include <string>
2 #include <vector>
3
4 class Foo {
5     std::string m_name;
6     std::vector<int> m_data;
7     std::size_t m_len;
8
9 public:
10     explicit Foo(const char *name)
11         : m_name{name}
12         , m_data{}
13         , m_len{this->m_name.length()}
14     {}
15 };
16
17 int main() {
18     Foo foo = Foo{"Rustikon"};
19
20     return 0;
21 }
```

# Manual memory initialization in Rust

**`std::mem::MaybeUninit`**



```
1 use std::mem::MaybeUninit;
2 use std::ptr;
3
4 pub struct Foo {
5     name: String,
6     data: Vec<i32>,
7     len: usize,
8 }
9
10 impl Foo {
11     pub fn new_in<'storage>(
12         this: &mut MaybeUninit<Self>,
13         name: &str
14     ) -> &'storage mut Self {
15         let this = this.as_mut_ptr();
16
17         let name = name.to_owned();
18         let data = Vec::new();
19
20         unsafe {
21             ptr::write(&raw mut (*this).name, name);
22             ptr::write(&raw mut (*this).data, data);
23
24             let len = unsafe {
25                 let name = &(&raw const (*this).name);
26                 name.len()
27             };
28             ptr::write(&raw mut (*this).len, len);
29         }
30
31         unsafe { &mut *this }
```

```
32     }
33 }
34
35 fn use_foo(_foo: &Foo) {}
36
37 fn main() {
38     let mut storage = MaybeUninit::uninit();
39
40     let foo = Foo::new_in(&mut storage, "Rustikon");
41     use_foo(&*foo);
42
43     unsafe {
44         storage.assume_init_drop();
45     }
46 }
```



# Manual memory initialization in C++

```
1  #include <string>
2  #include <vector>
3
4  class Foo {
5      std::string m_name;
6      std::vector<int> m_data;
7      std::size_t m_len;
8
9  public:
10     explicit Foo(const char *name)
11         : m_name{name},
12           m_data{},
13           m_len{this->m_name.length()}
14     {}
15 };
16
17 void use_foo(const Foo &foo) {}
```

```
18
19 int main() {
20     alignas(Foo) std::byte
21     buf[sizeof(Foo)];
22
23     Foo *foo = new (buf)
24     Foo{"Rustikon"};
25     use_foo(*foo);
26     foo->~Foo();
27 }
```

# Destructors, Drop, and Drop-glue

# Normal memory deinitialization in C++



```
1  #include <print>
2  #include <string>
3
4  class Logger {
5      std::string m_message;
6
7  public:
8      Logger(const char *message) :
9          m_message{message} {}
10     ~Logger() { std::println!("{}", m_message); }
11 };
12
13 class Foo {
14     Logger a;
15     Logger b;
16     Logger c;
17
18 public:
19     Foo() : a{"a"}, b{"b"}, c{"c"} {}
20     ~Foo() { std::println("Foo"); }
21 };
22
```

```
22 int main() {
23     Foo foo = Foo{};
24
25     return 0;
26 }
```

```
1 Output:
2 Foo
3 c
4 b
5 a
```

# Normal memory deinitialization in Rust



```
1 pub struct Logger(String);
2
3 impl Drop for Logger {
4     fn drop(&mut self) {
5         println!("{}", self.0);
6     }
7 }
8
9 pub struct Foo {
10     a: Logger,
11     b: Logger,
12     c: Logger,
13 }
14
15 impl Drop for Foo {
16     fn drop(&mut self) {
17         println!("Foo");
18     }
19 }
20
21 impl Foo {
22     pub fn new() -> Self {
```

```
23         Self {
24             a: Logger("a".to_owned()),
25             b: Logger("b".to_owned()),
26             c: Logger("c".to_owned()),
27         }
28     }
29 }
30
31 fn main() {
32     let foo = Foo::new();
33 }
```

```
1 Output:
2 Foo
3 a
4 b
5 c
```

# Controlling drop order



- 1. Disable drop-glue**
- 2. Run destructors manually**

**std::mem::ManuallyDrop**



```
1 use std::mem::ManuallyDrop;
2
3 pub struct Logger(String);
4
5 impl Drop for Logger {
6     fn drop(&mut self) {
7         println!("{}", self.0);
8     }
9 }
10
11 struct Foo {
12     a: ManuallyDrop<Logger>,
13     b: ManuallyDrop<Logger>,
14     c: ManuallyDrop<Logger>,
15 }
16
17 impl Drop for Foo {
18     fn drop(&mut self) {
19         unsafe {
20             ManuallyDrop::drop(&mut self.a);
21             ManuallyDrop::drop(&mut self.c);
22             ManuallyDrop::drop(&mut self.b);
23         }
24     }
25 }
26
```

```
27 impl Foo {
28     fn new() -> Self {
29         Self {
30             a:
31             ManuallyDrop::new(Logger("a".to_owned())),
32             b:
33             ManuallyDrop::new(Logger("b".to_owned())),
34             c:
35             ManuallyDrop::new(Logger("c".to_owned())),
36         }
37     }
38 }
39
40 fn main() {
41     let foo = Foo::new();
42 }
```

## 1 Output:

```
2 a
3 c
4 b
```

C++

# With runtime overhead:

- `std::optional`
- `std::unique_ptr`

**union**



```
1  #include <print>
2  #include <string>
3
4  class Logger {
5      std::string m_message;
6
7  public:
8      Logger(const char *message) :
9          m_message{message} {}
10     ~Logger() { std::println("{} ", m_message); }
11 };
12
13 class Foo {
14     union {
15         Logger a;
16     };
17     union {
18         Logger b;
19     };
20     union {
21         Logger c;
22     };
23 }
```

```
22
23 public:
24     Foo() : a{"a"}, b{"b"}, c{"c"} {}
25     ~Foo() {
26         a.~Logger();
27         c.~Logger();
28         b.~Logger();
29     }
30 };
31
32 int main() {
33     Foo foo = Foo{};
34
35     return 0;
36 }
```

1 Output:

```
2 a
3 c
4 b
```

# Summary



|                               | Rust                        | C++                                        |
|-------------------------------|-----------------------------|--------------------------------------------|
| initialization order          | unspecified                 | In the order of declaration                |
| in-place initialization       | Manual. Open design space   | Yes. Placement new                         |
| destruction order             | In the order of declaration | In the <i>reverse</i> order of declaration |
| controlling destruction order | ManuallyDrop                | union                                      |



Public repository with slides